01 / PRODUCT OVERVIEW

MNC DIAG-01

Automotive diagnostic interface | Product development

UNDERSTAND THE VEHICLE

MNC DIAG-01 is a diagnostic interface in development at MENOCLONE Motorsports. The project explores a practical way to read supported vehicle data and present it in a clear, useful format.

Today's vehicles depend on electronic control modules, sensors, and communication networks. Reading those systems can help drivers and technicians understand a problem before deciding what to inspect or repair.

PROJECT FOCUS

The first goal is a working laboratory prototype. Future versions may combine onboard electronics, diagnostic software, and a straightforward display of vehicle information.

Development status: prototype research and bench testing. No production release or vehicle compatibility list has been confirmed.

02 / BOARD ARCHITECTURE

Inside DIAG-01

The hardware being evaluated for the first prototype

The development setup uses an **ESP32-S3** microcontroller, a compatible **CAN transceiver**, and a separate USB CAN adapter for testing. Early communication tests take place on an isolated CAN network before any vehicle connection is considered.

HOW THE SYSTEM WORKS

OBD-II connection
Provides a physical interface to supported vehicle diagnostic systems.

CAN transceiver
Handles the electrical signaling between the CAN network and the controller.

ESP32-S3 controller
Runs firmware to send requests and process supported responses.

Computer or display
Shows decoded information for testing and later software development.

The final PCB design will require validated power protection, CAN protection, connector pinout, and thermal performance.

03 / DIAGNOSTIC FUNCTIONS

Reading the signals

Initial software and diagnostic development

The first laboratory objective is to send simulated OBD-II requests and decode responses such as engine RPM. Later development may add supported trouble codes, live readings, readiness status, and stored diagnostic information.

A TYPICAL DIAGNOSTIC SESSION

01 Connect / Establish communication using verified hardware and a supported protocol.

02 Read / Request available data and diagnostic trouble codes.

03 Record / Save relevant codes, operating conditions, and live readings.

04 Diagnose / Compare results with vehicle-specific service information.

05 Verify / After repairs, confirm the fault has been addressed.

A fault code is a starting point, not a parts replacement instruction. Access to manufacturer-specific systems such as ABS, airbags, and body modules may require other hardware or software.

04 / BUILD PLAN

Development roadmap

A measured approach from experiments to a custom board

01 / CONTROLLER SETUP

Program the ESP32-S3 from Ubuntu and confirm stable USB serial communication.

02 / CAN BENCH NETWORK

Connect two isolated CAN nodes with correct termination and verify message transfer.

03 / DATA DISPLAY

Decode simulated requests and show results on a computer. Evaluate a future Auto Insights interface.

04 / PCB DEVELOPMENT

Design and review the custom circuit, then build and test an assembled prototype.

Electrical safety: Never apply vehicle 12 V directly to the ESP32-S3. Vehicle testing must wait until automotive power protection, grounding, connector wiring, and CAN circuitry have been checked.

05 / THE LARGER VISION

Built to keep evolving

MENOCLONE Motorsports | Automotive electronics and community

DIAG-01 is one part of a growing effort to combine electronics, software, and automotive learning. MENOCLONE Motorsports is developing products and concepts with the everyday driver in mind, while documenting the work that goes into each stage.

CONNECTED PROJECTS

MNC UNO R1 explores circuit and microcontroller fundamentals. **MNC DIAG-01** applies those skills to automotive communication. **Auto Insights** is the proposed software interface for displaying supported diagnostic readings.

The long-term goal is to build useful tools, share engineering knowledge, and involve the MENOCLONE Auto Club community in the process of learning and improving.

ORIGINALITY HAS ORIGINS.

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